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Digital Circuits Lab

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Objective:

Designing and simulating a 4-bit counter with 7-segment displays and four flip flops and a push button as a manual clock.

Details:

1. To begin, open Logisim and construct a four-bit counter using a push button, four D-flip flops, an IC 7447 seven segment decoder chip, inverters, and a 7-segment display.
2. To complete this ensure the outputs of each D-flipflop is hooked up to the decoder chip (ie. Q0-A, Q1-B, etc.) and that the decoder chip outputs are hooked up properly to the 7-segment display
3. The completed gate level 4-bit counter should appear similar to Figure 1.
4. Recreate this same circuit on a chip-level using two IC 7474 chips, the same 7447 decoder chip, two IC 7404 chips, and a 7-segment display
5. The completed chip-level schematic should appear similar to Figure 2.
6. Test both circuits and record their outputs on a truth table.
7. Next simulate an 8-bit counter using the same components as the previous 4-bit counter on both the gate and chip level. (See Figures 3 and 4).
8. Next, using the 4-bit counter create a circuit using a multiplexer that when a switch is flipped the display of the counter is shifted between two 7-segment displays.
9. Do this on both a gate and chip level, in the simulation you may represent which display is on by using the decimal pin on the 7-segment display.
10. Your completed circuit should appear similar to Figures 5 and 6.
11. Record the outputs of your circuits and record them in a truth table.
12. To construct this circuit in person, rather than using a multiplexer you will need an IC 74257 chip to multiplex which display is grounded so that when the switch is flipped the display that is enabled will change.
13. You will need each chip used in Logisim to create the multiplexer schematic on the protoboard.
14. When constructing on protoboard ensure all chips are powered and grounded, the board has the necessary power connections, so the buttons and switches work as well as the D-flipflop chips have power on their low voltage function pins to ensure the counters are working correctly.
15. Your circuit should appear similar to Figure 7, record its outputs and take a video of your circuit.

Table 1: Outputs of 4-Bit D-Flipflop Counters

D-Flipflop Outputs				Number shown on 7- Segment Display
Q0	Q1	Q2	Q3	
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9

Table 2: Output of Multiplexed 4-Bit Counters

D-Flipflop Outputs				Simulated			Experiemental		
Q0	Q1	Q2	Q3	Switch	Number Shown on First 7-Segment Display	Number Shown on Second 7-Segment Display	S0	DIG0	DIG1
0	0	0	0	0	0	N/A	0	0	N/A
0	0	0	1	0	1	N/A	0	1	N/A
0	0	1	0	0	2	N/A	0	2	N/A
0	0	1	1	0	3	N/A	0	3	N/A
0	1	0	0	0	4	N/A	0	4	N/A
0	1	0	1	0	5	N/A	0	5	N/A
0	1	1	0	0	6	N/A	0	6	N/A
0	1	1	1	0	7	N/A	0	7	N/A
1	0	0	0	0	8	N/A	0	8	N/A
1	0	0	1	0	9	N/A	0	9	N/A
0	0	0	0	1	N/A	0	1	N/A	0
0	0	0	1	1	N/A	1	1	N/A	1
0	0	1	0	1	N/A	2	1	N/A	2
0	0	1	1	1	N/A	3	1	N/A	3
0	1	0	0	1	N/A	4	1	N/A	4
0	1	0	1	1	N/A	5	1	N/A	5
0	1	1	0	1	N/A	6	1	N/A	6
0	1	1	1	1	N/A	7	1	N/A	7
1	0	0	0	1	N/A	8	1	N/A	8
1	0	0	1	1	N/A	9	1	N/A	9

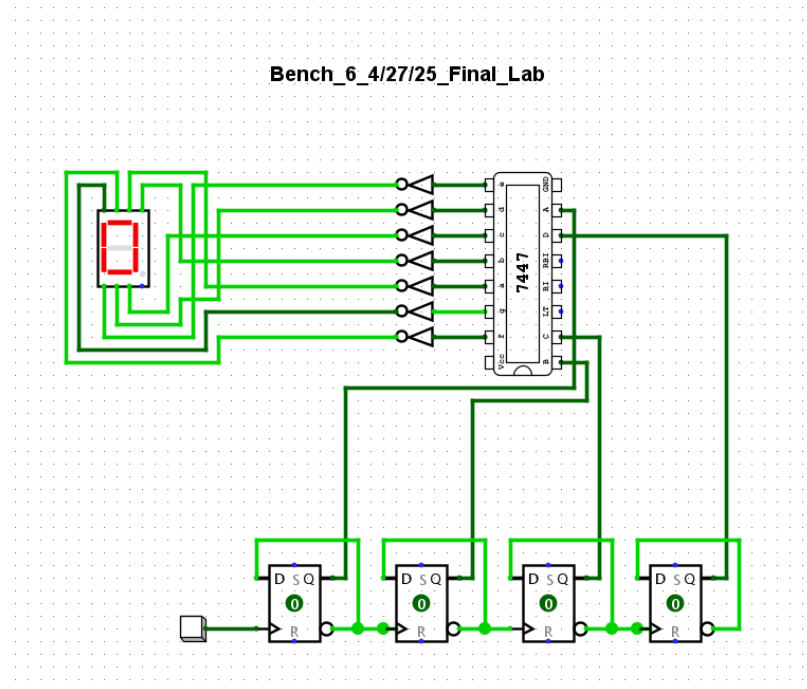


Figure 1: Gate Level D-Flipflop 4-Bit Counter

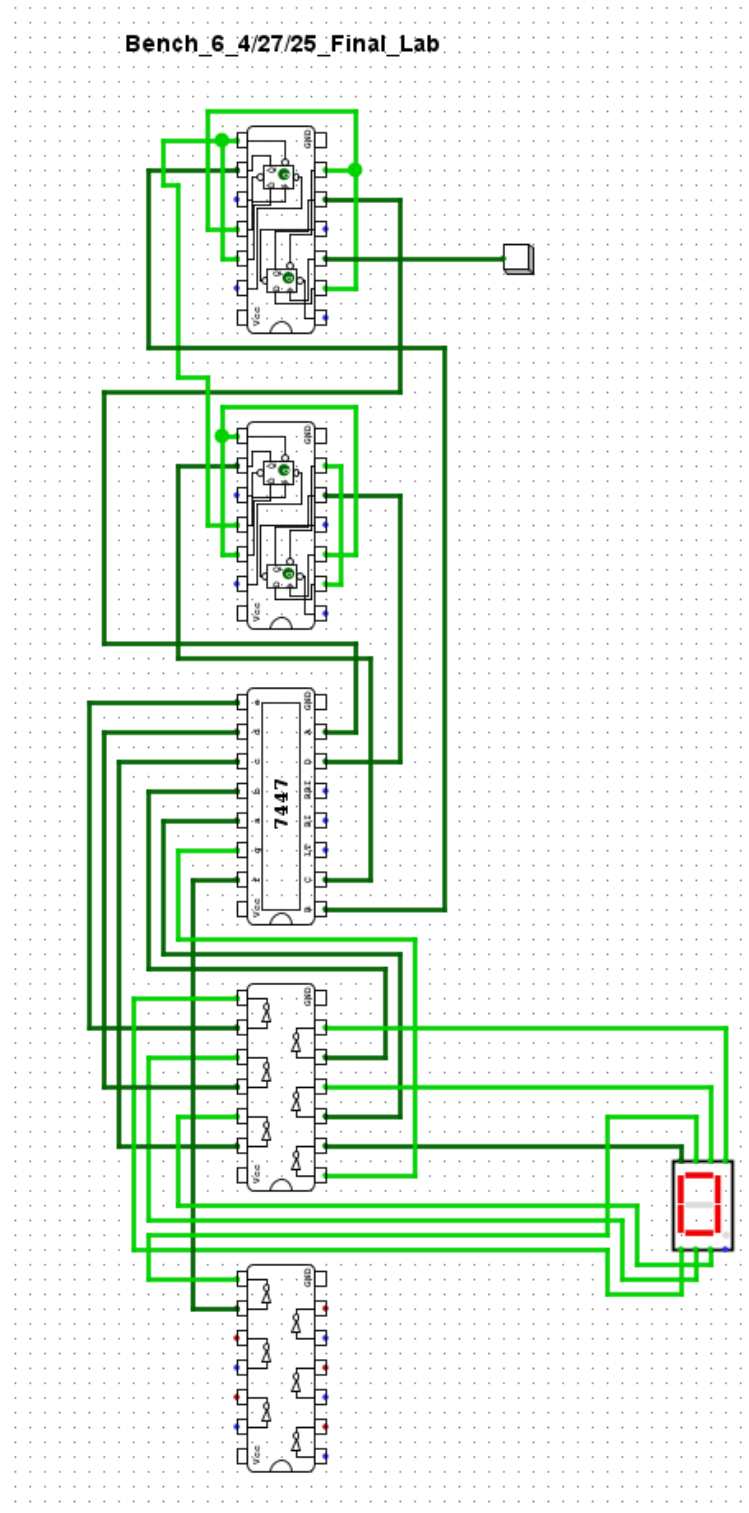


Figure 2: Completed Chip-Level D-Flipflop 4-Bit Counter

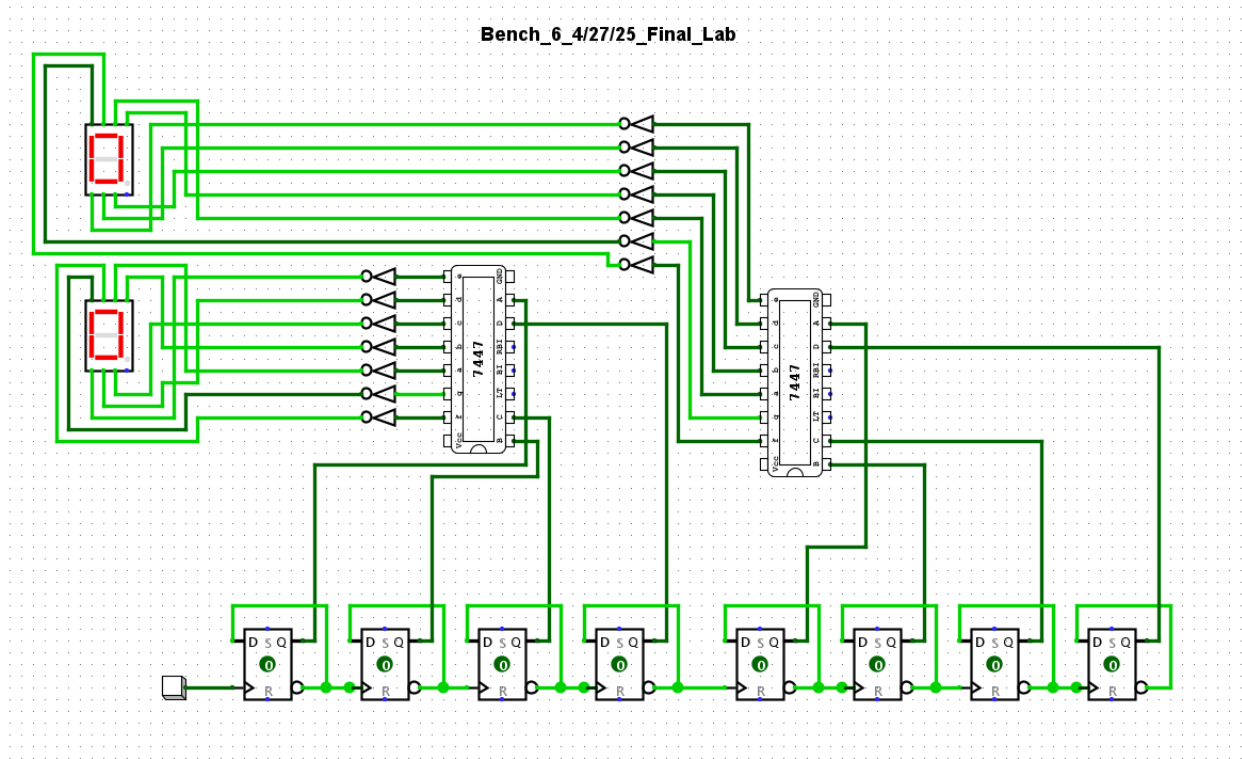


Figure 3: Completed Gate-Level 8-Bit Counter

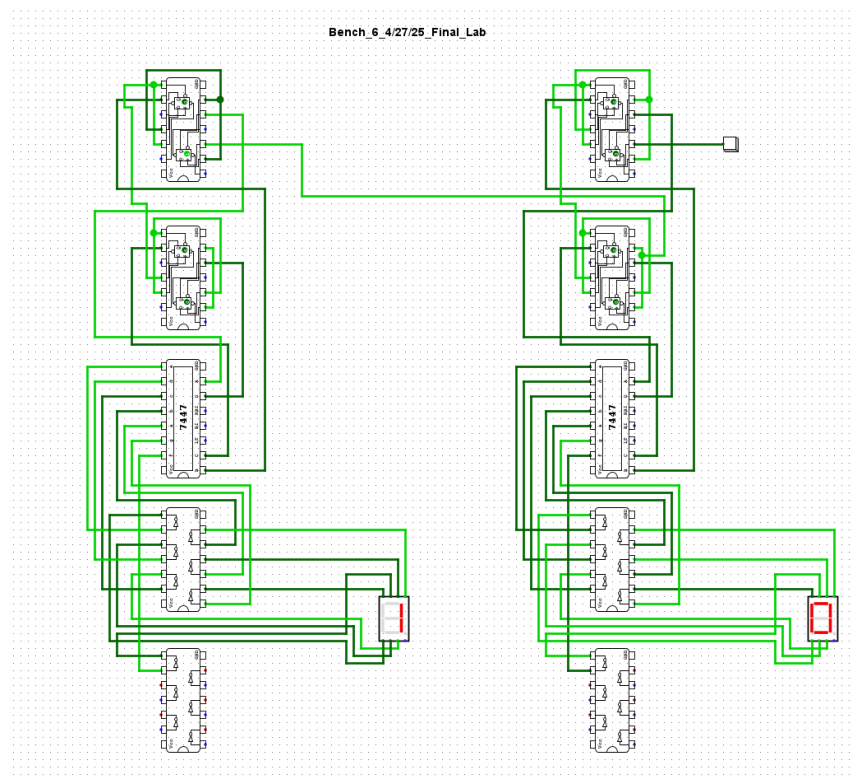


Figure 4: Completed Chip-Level 8-Bit Counter

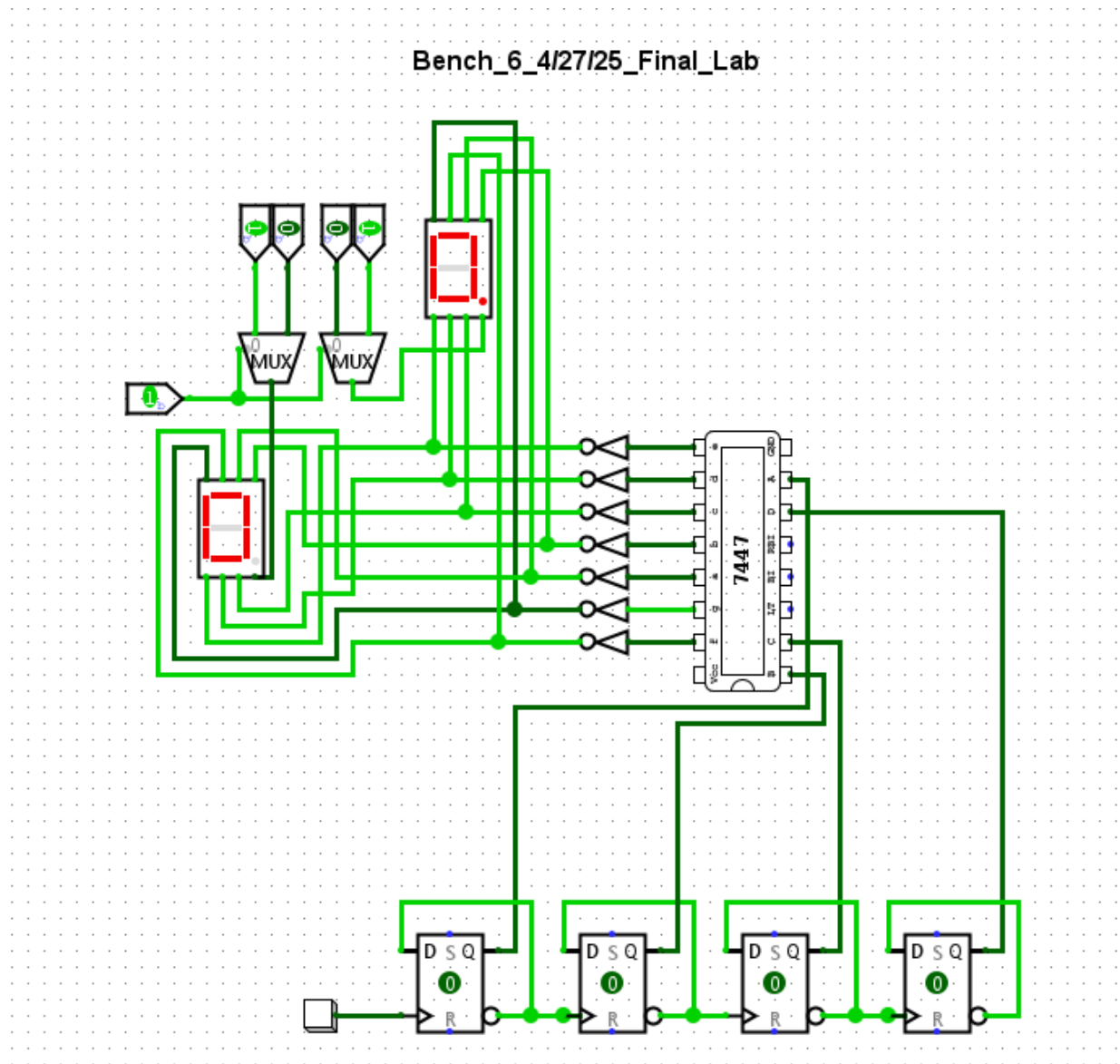


Figure 5: Completed Gate-Level 4-Bit Counter with Multiplexed Displays

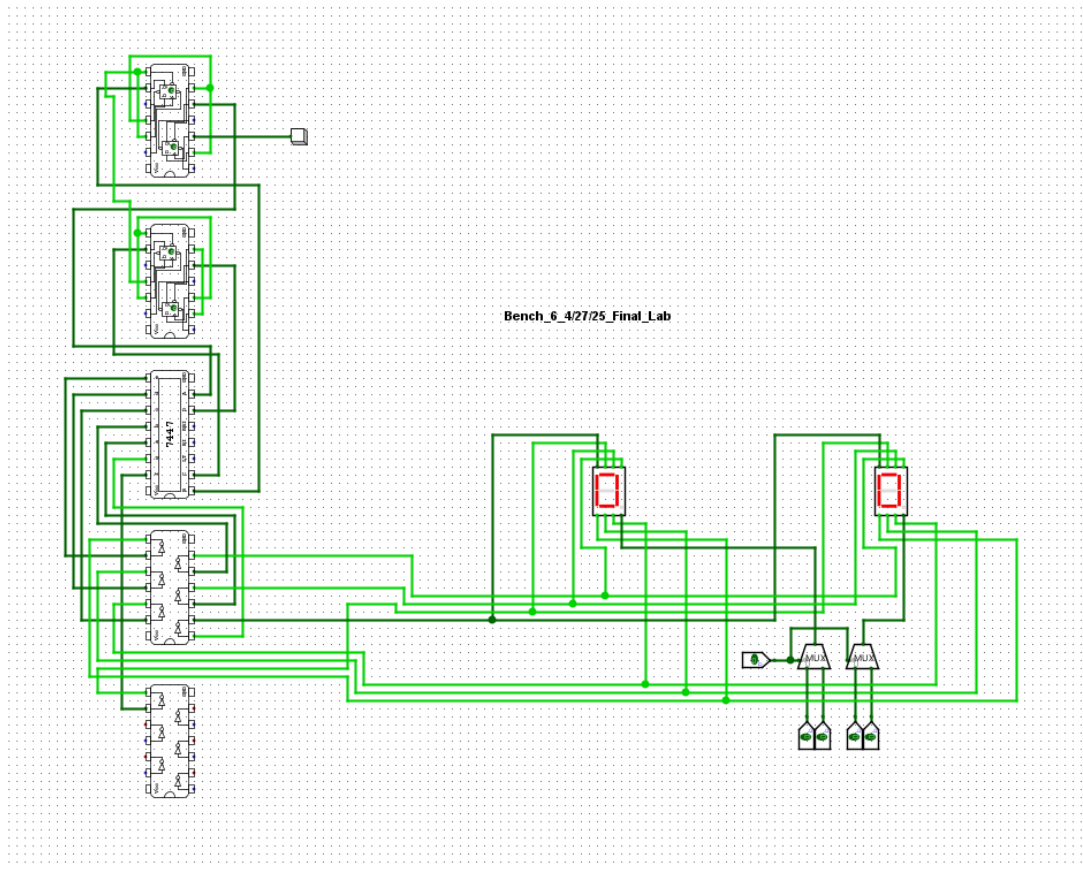


Figure 6: Completed Chip-Level 4-Bit Counter with Multiplexed Displays

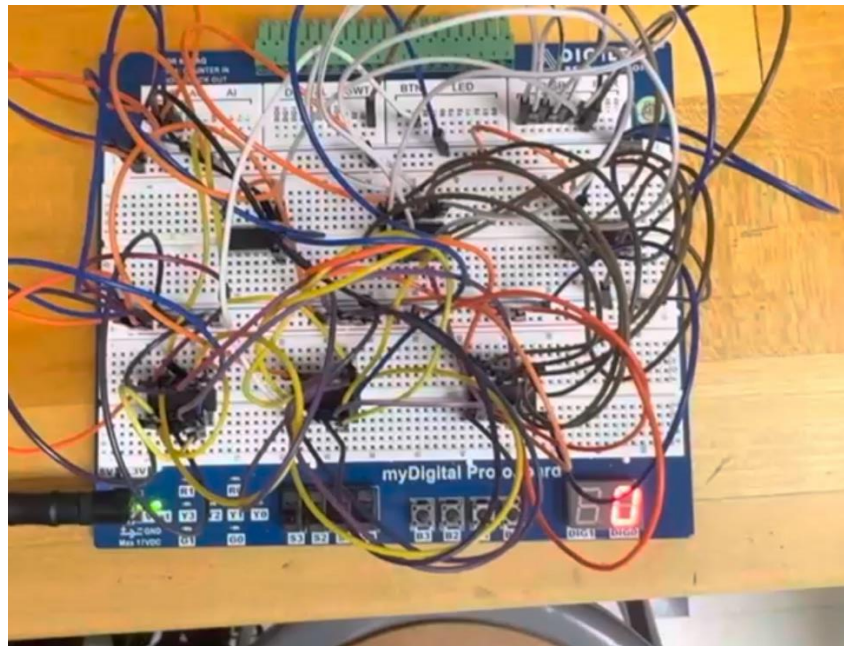


Figure 7: Completed 4-Bit Counter with Multiplexed Displays on Protoboard

Conclusion:

This lab had challenged me with understanding the function of a multiplexer, especially at the chip level, I feel I have learned how to utilize it and understand it better through this lab. Our design process was slow as we had issues with the Logisim software when creating our 4-bit counter but after finding the solution to our issue we were able to construct each schematic with ease. I am thankful our wiring went smoothly; we had initially forgotten to disable the low-voltage functions of the IC 7474 chip, so we had some error outputs but after sorting those out we had completed our circuit with ease. I feel this lab, through each circuit design, helped me better understand the applications of logic gates in a real world setting and how I can utilize them in my career.